# Operating Systems: Assignment-2

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Q 1. Write a Shell Script to find factorial of a number.

**Code:**

```
#!/bin/bash

echo "Enter a number:"
read n

fact=1

if [ $n -lt 0 ]; then
  echo "Factorial is not defined for negative numbers"
  exit 1
fi

for (( i=1; i<=n; i++ ))
do
  fact=$((fact * i))
done

echo "Factorial of $n is: $fact"
```

**Output:**

```
Enter a number:
7
./1.sh: line 8: [7: command not found
Factorial of 7 is: 5040
```

Q 2. Write a shell script for a program to implement an address book with the options given below:

a) Create an address book.
b) View the address book.
c) Insert a record.
d) Delete a record.
e) Modify a record.
f) Exit.

**Code:**

```
#!/bin/bash

BOOK="addressbook.txt"

while true
do
  echo "ADDRESS BOOK"
  echo "1. Create Address Book"
  echo "2. View Address Book"
  echo "3. Insert a Record"
  echo "4. Delete a Record"
  echo "5. Modify a Record"
  echo "6. Exit"
  echo "Enter your choice:"
  read choice

  case $choice in

  1)
    if [ -f $BOOK ]; then
      echo "Address book already exists!"
    else
      touch $BOOK
      echo "Address book created successfully."
    fi
    ;;

  2)
    if [ -s $BOOK ]; then
      echo "Address Book Records"
```

```
        cat $BOOK
    else
        echo "Address book is empty or does not exist."
    fi
    ;;

3)
    echo "Enter Name:"
    read name
    echo "Enter Phone:"
    read phone
    echo "Enter Email:"
    read email
    echo "Enter Address:"
    read address

    echo "$name | $phone | $email | $address" >> $BOOK
    echo "Record inserted successfully."
    ;;

4)
    echo "Enter name to delete record:"
    read name
    if grep -q "^$name" $BOOK; then
        grep -v "^$name" $BOOK > temp.txt
        mv temp.txt $BOOK
        echo "Record deleted successfully."
    else
        echo "Record not found."
    fi
    ;;

5)
    echo "Enter name to modify record:"
    read name
    if grep -q "^$name" $BOOK; then
        echo "Enter new Phone:"
        read phone
        echo "Enter new Email:"
```

```
            read email
            echo "Enter new Address:"
            read address

            sed -i "/^$name/d" $BOOK
            echo "$name | $phone | $email | $address" >> $BOOK
            echo "Record modified successfully."
        else
            echo "Record not found."
        fi
        ;;

    6)
        echo "Exiting Address Book..."
        exit 0
        ;;

    *)
        echo "Invalid choice. Please try again."
        ;;
    esac
done
```

**Output:**

```
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
4. Delete a record
5. Modify a record
6. Exit
Enter your choice:
1
Address book already exists!
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
4. Delete a record
5. Modify a record
6. Exit
Enter your choice:
3
Enter Name:
Leon
Enter Phone:
9876543210
Enter Email:
leon@gmail.com
Enter Address:
SVB Bhavan
Record inserted successfully
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
```

```
4. Delete a record
5. Modify a record
6. Exit
Enter your choice:
2
Address Book Records
Leon | 9876543210 | leon@gmail.com | SVB Bhavan
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
4. Delete a record
5. Modify a record
6. Exit
Enter your choice:
5
Enter name to modify record:
Leon
Enter new Phone
1234567891
Enter new Email:
lobo@gmail.com
Enter new Address:
same
Record modified successfully.
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
4. Delete a record
5. Modify a record
6. Exit
```

```
Enter your choice:
4
Enter name to delete record:
Leon
Record deleted successfully.
ADDRESS BOOK
1. Create Address Book
2. View Address Book
3. Insert a record
4. Delete a record
5. Modify a record
6. Exit
Enter your choice:
6
Exiting addressBook...
```

Q 3. Write the shell script that renames all files in the current directory that end in ".jpg" to begin with today's date in the following format: YYYY-MM-DD. For example, if a picture of cat was in the current directory and today was October 31,2016 it would change name from "mycat.jpg" to "2016–10–31-mycat.jpg"

**Code:**

```
#!/bin/bash

date_prefix=$(date +"%Y-%m-%d")

for file in *.jpg
do
  [ -e "$file" ] || continue

  mv "$file" "$date_prefix-$file"
done

echo "All .jpg files renamed successfully."
```

**Output:**

```
All .jpg files renamed successfully.
ubuntu@ubuntu:~/test$ ls
1.sh                  addressbook.sh   factorial.sh  rename_jpg.sh
2026-01-22-mycat.jpg  addressbook.txt  file.txt
```

Q 4. Write a shell script for a program with options to calculate the following:

a) Area of a circle

b) Circumference of a circle

c) Area of a rectangle

d) Area of a square

**Code:**

```
#!/bin/bash

while true
do
   echo "GEOMETRY MENU"
   echo "1. Area of a Circle"
   echo "2. Circumference of a Circle"
   echo "3. Area of a Rectangle"
   echo "4. Area of a Square"
   echo "5. Exit"
   echo "Enter your choice:"
   read choice

   case $choice in

   1)
      echo "Enter radius:"
      read r
      area=$(echo "3.14 * $r * $r" | bc)
      echo "Area of Circle = $area"
      ;;

   2)
      echo "Enter radius:"
      read r
      circ=$(echo "2 * 3.14 * $r" | bc)
      echo "Circumference of Circle = $circ"
      ;;

   3)
      echo "Enter length:"
      read l
```

```
        echo "Enter breadth:"
        read b
        area=$(echo "$l * $b" | bc)
        echo "Area of Rectangle = $area"
        ;;

    4)
        echo "Enter side:"
        read s
        area=$(echo "$s * $s" | bc)
        echo "Area of Square = $area"
        ;;

    5)
        echo "Exiting program..."
        exit 0
        ;;

    *)
        echo "Invalid choice! Try again."
        ;;
    esac
done
```

**Output:**

```
GEOMETRY MENU
1. Area of a Circle
2. Circumference of a Circle
3. Area of a Rectangle
4. Area of a Square
5. Exit
Enter your choice:
1
Enter radius:
12
Area of Circle = 452.16
GEOMETRY MENU
1. Area of a Circle
2. Circumference of a Circle
3. Area of a Rectangle
4. Area of a Square
5. Exit
Enter your choice:
2
Enter radius:
12
Circumference of Circle = 75.36
GEOMETRY MENU
1. Area of a Circle
2. Circumference of a Circle
3. Area of a Rectangle
4. Area of a Square
5. Exit
Enter your choice:
3
```

```
Enter length:
13
Enter breadth:
3
Area of Rectangle = 39
GEOMETRY MENU
1. Area of a Circle
2. Circumference of a Circle
3. Area of a Rectangle
4. Area of a Square
5. Exit
Enter your choice:
4
Enter side:
5
Area of Square = 25
GEOMETRY MENU
1. Area of a Circle
2. Circumference of a Circle
3. Area of a Rectangle
4. Area of a Square
5. Exit
Enter your choice:
5
Exiting program...
```